1. Software Design Specification

HYBRID_LIB_ADA

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Author(s)	Michael Gardner
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Project Profile	
Variant	library
Library Role	enterprise
Application Role	N/A
Assurance	spark-targeted
Execution	sequential
Parallelism	none
Platform	desktop, server, embedded
Execution Environment	linux, macos, windows, ada-runtime
Processor Architecture	amd64, arm64, arm32
Deployment	native

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2. Introduction

2.1. Purpose

This Software Design Specification (SDS) describes the internal architecture, package structure, key type definitions, build strategy, and design decisions for **HYBRID_LIB_ADA**.

2.2. Scope

This document covers:

- the 4-layer hybrid library architecture,
- the three-package API pattern,
- package hierarchy and dependency rules,
- core type definitions and contracts,
- static dependency injection via Ada generics,
- SPARK verification boundaries, and
- the layered exception-boundary enforcement strategy.

2.3. References

- Software Requirements Specification (SRS).
- `docs/guides/all_about_our_api.md`.
- `docs/guides/architecture_enforcement.md`.
- Supporting UML diagrams maintained in `docs/diagrams/`.
- Ada 2022 Reference Manual.
- SPARK Reference Manual.

3. Architectural Overview

3.1. Layer Architecture

HYBRID_LIB_ADA uses the shared hybrid architecture family and the same inner core as the application variant: **Domain → Application → Infrastructure**. The library profile differs only at the outer boundary, where it uses the API pattern instead of Presentation + Bootstrap.

Layer / Boundary	Purpose	Depends On
API Facade (`Hybrid_Lib_Ada.API`)	Thin public facade; re-exports public types and delegates operations to the default composition root.	Application + Domain
API Composition Root (`API.Desktop`)	Default desktop/server composition root; wires infrastructure adapters into the generic operations surface.	All layers
API Operations (`API.Operations`)	SPARK-safe operations surface parameterized by port implementations.	Application + Domain
Infrastructure	Adapters for I/O and external integration; implements application-owned ports.	Application + Domain
Application	Use cases, commands, and port definitions.	Domain

Layer / Boundary	Purpose	Depends On
Domain	Pure business concepts, value objects, and error types.	Nothing

Detailed architecture guidance is maintained in `docs/guides/`. Supporting UML diagrams are maintained in `docs/diagrams/`.

3.2. Dependency Rules

The API layer has **two distinct areas** with different dependency rules.

Component	May Depend On
Domain	Nothing (zero dependencies).
Application	Domain only.
Infrastructure	Application and Domain.
API facade (`api/`)	Application + Domain only.
API composition roots (`api/desktop/`)	All layers, including Infrastructure.

The API facade (`Hybrid_Lib_Ada.API`, `API.Operations`) provides the public interface and does not import Infrastructure. The composition root (`API.Desktop`) is the intentional exception that wires concrete adapters to the generic operations surface.

Architecture conformance is enforced through layered controls:
- project/build configuration constraints where applicable,
- active validation using `scripts/arch_guard.py`,
- execution of `scripts/arch_guard.py` during normal developer build workflow, and
- execution of `scripts/arch_guard.py` in CI/CD before non-conforming changes are accepted.

3.3. Hexagonal Pattern

The Application layer defines ports and use cases. Infrastructure adapters implement outbound ports. Domain and Application remain independent of concrete I/O concerns.

Element	Role
Application.Port.Outbound.Writer	Application-owned outbound port contract.
Application.Usecase.Greet	Orchestrates the greeting use case using domain validation and an outbound Writer port.
Infrastructure.Adapter.Console_Writer	Concrete infrastructure adapter implementing the Writer port.

4. Package Structure

4.1. Directory Layout

```
src/
├── hybrid_lib_ada.ads              # Root package
├── version/
│   └── hybrid_lib_ada-version.ads  # Version information
│
├── domain/
│   ├── domain.ads                  # Domain layer root
│   ├── domain-unit.ads             # Unit type
```

```
│   ├── error/
│   │   ├── domain-error.ads        # Error type definition
│   │   └── domain-error-result.ads # Generic Result monad
│   └── value_object/
│       ├── domain-value_object.ads
│       ├── domain-value_object-option.ads
│       └── domain-value_object-person.ads
│
├── application/
│   ├── application.ads
│   ├── error/
│   │   └── application-error.ads   # Re-exports Domain.Error
│   ├── command/
│   │   ├── application-command.ads
│   │   └── application-command-greet.ads
│   ├── port/
│   │   ├── application-port.ads
│   │   ├── inbound/
│   │   │   └── application-port-inbound.ads
│   │   └── outbound/
│   │       ├── application-port-outbound.ads
│   │       └── application-port-outbound-writer.ads
│   └── usecase/
│       ├── application-usecase.ads
│       └── application-usecase-greet.ads
│
├── infrastructure/
│   ├── infrastructure.ads
│   └── adapter/
│       ├── infrastructure-adapter.ads
│       └── infrastructure-adapter-console_writer.ads
│
└── api/
    ├── hybrid_lib_ada-api.ads
    ├── hybrid_lib_ada-api.adb
    ├── operations/
    │   ├── hybrid_lib_ada-api-operations.ads
    │   └── hybrid_lib_ada-api-operations.adb
    └── desktop/
        ├── hybrid_lib_ada-api-desktop.ads
        └── hybrid_lib_ada-api-desktop.adb
```

4.2. Package Descriptions

4.2.1. Domain Layer

Package	Purpose	SPARK
`Domain`	Layer root.	On
`Domain.Error`	Error type with kind and bounded message.	On
`Domain.Error.Result`	Generic Result[T] monad.	On
`Domain.Unit`	Unit type for void-like operations.	On
`Domain.Value_Object`	Value object root.	On
`Domain.Value_Object.Option`	Option[T] monad.	On
`Domain.Value_Object.Person`	Person value object.	On

4.2.2. Application Layer

Package	Purpose	SPARK
`Application`	Layer root.	On
`Application.Command`	Command root.	On
`Application.Command.Greet`	Greeting command DTO.	On
`Application.Error`	Re-exports Domain.Error.	On
`Application.Port`	Port root.	On
`Application.Port.Outbound.Writer`	Writer port definition.	On
`Application.Usecase.Greet`	Greeting use case.	On

4.2.3. Infrastructure Layer

Package	Purpose	SPARK
`Infrastructure`	Layer root.	Off
`Infrastructure.Adapter`	Adapter root.	Off
`Infrastructure.Adapter.Console_Writer`	Console output adapter.	Off

4.2.4. API Layer

Package	Purpose	SPARK
`Hybrid_Lib_Ada`	Library root.	Off
`Hybrid_Lib_Ada.API`	Public facade.	Off
`Hybrid_Lib_Ada.API.Desktop`	Desktop composition root.	Off
`Hybrid_Lib_Ada.API.Operations`	SPARK-safe operations surface.	On
`Hybrid_Lib_Ada.Version`	Version information.	Off

5. Type Definitions

5.1. Domain Types

5.1.1. Error_Kind

```
type Error_Kind is
  (Validation_Error,
   Parse_Error,
   Not_Found_Error,
   IO_Error,
   Internal_Error);
```

5.1.2. Error_Type

```
type Error_Type is record
   Kind    : Error_Kind;
   Message : Error_Strings.Bounded_String;  -- Max 512 chars
end record;
```

5.1.3. Result (Generic)

```
generic
   type T is private;
package Domain.Error.Result.Generic_Result is
   type Result is private;
```

```
   function Ok (Value : T) return Result;
   function Error (Kind : Error_Kind; Message : String) return Result;

   function Is_Ok (R : Result) return Boolean;
   function Is_Error (R : Result) return Boolean;
   function Value (R : Result) return T with Pre => Is_Ok (R);
   function Error_Info (R : Result) return Error_Type with Pre => Is_Error (R);
end Generic_Result;
```

5.1.4. Person

```
type Person is private;

function Create (Name : String) return Person_Result.Result;
function Get_Name (P : Person) return String;
function Is_Valid_Person (P : Person) return Boolean;
```

5.2. Application Types

5.2.1. Greet_Command

```
type Greet_Command is private;

function Create (Name : String) return Greet_Command;
function Get_Name (Cmd : Greet_Command) return String;
```

5.2.2. Writer Port

```
generic
   with function Write (Message : String) return Unit_Result.Result;
package Generic_Writer is
   function Write_Message (Message : String) return Unit_Result.Result
   renames Write;
end Generic_Writer;
```

5.3. API Types

All public types are re-exported from `Hybrid_Lib_Ada.API`.

```
subtype Person_Type is Domain.Value_Object.Person.Person;
subtype Greet_Command is Application.Command.Greet.Greet_Command;
subtype Error_Type is Domain.Error.Error_Type;
subtype Error_Kind is Domain.Error.Error_Kind;
```

6. Design Patterns

6.1. Static Dependency Injection

HYBRID_LIB_ADA uses Ada generics for static (compile-time) dependency injection.

```
-- 1. Port defines generic signature
package Generic_Writer is ...

-- 2. Use case or operations package is generic, parameterized by Writer
package Application.Usecase.Greet is ...

-- 3. Composition root instantiates with concrete adapter
package Console_Ops is new Hybrid_Lib_Ada.API.Operations
  (Writer => Infrastructure.Adapter.Console_Writer.Write);
```

Benefit	Description
SPARK compatible	No runtime dispatching is required.
Testable	Mock writers and alternate composition roots are easy to provide.
Type safe	The compiler verifies contracts and instantiations.
Zero runtime overhead	Monomorphization occurs at compile time.

6.2. Three-Package API Pattern

The library profile uses a three-package API pattern:

- `Hybrid_Lib_Ada.API` as the public facade,
- `Hybrid_Lib_Ada.API.Desktop` as the default composition root,
- `Hybrid_Lib_Ada.API.Operations` as the SPARK-safe operations surface.

Package	Design Role	SPARK
`Hybrid_Lib_Ada.API`	Thin public facade; re-exports public types and delegates to the default composition root.	Off
`Hybrid_Lib_Ada.API.Desktop`	Composition root for desktop/server environments; wires adapters into operations.	Off
`Hybrid_Lib_Ada.API.Operations`	Generic operations surface parameterized by outbound ports; safe verification boundary.	On

6.3. Result Monad Pattern

All fallible operations return `Result[T]` values rather than propagating expected failures as exceptions.

```
function Create (Name : String) return Person_Result.Result;
-- Returns Ok(Person) or Error(Validation_Error, "message")

function Greet (Cmd : Greet_Command) return Unit_Result.Result;
-- Returns Ok(Unit) or Error(IO_Error, "message")
```

7. Error Handling Strategy

7.1. Error Propagation

Errors flow through use-case orchestration as Result values.

```
function Execute (Cmd : Greet_Command) return Unit_Result.Result is
   Person_Res : constant Person_Result.Result := Person.Create (Get_Name (Cmd));
begin
   if Person_Result.Is_Error (Person_Res) then
      return Unit_Result.Error (...);
   end if;

   return Writer (Format_Greeting (Person_Result.Value (Person_Res)));
end Execute;
```

7.2. No Exceptions Policy

Situation	Handling
I/O failure	Return Error result.

Situation	Handling
Programmer error in core layers	Fail loudly; treat as a bug to fix rather than a normal runtime condition.
Unexpected error at infrastructure boundary	Convert to `Internal_Error` result at the boundary.
Validation failure	Return Error result.

7.3. Exception Boundary Specification

This project uses a layered exception-boundary policy.

Layer	Functional.Try	Manual `exception`	Rationale
API	N/A	Forbidden	Public facade; returns Result values to consumers.
Application	N/A	Forbidden	Use-case orchestration; works with Result values only.
Domain	N/A	Forbidden	Pure core logic; works with Result values only.
Infrastructure	Required at external boundaries	Forbidden	Must convert external exceptions to Result values consistently.
Test	Optional	Allowed	Tests may deliberately verify exception scenarios.

7.3.1. Required Pattern: Functional.Try at Boundaries

All Infrastructure adapters that perform I/O or call external APIs shall use `Functional.Try.Map_To_Result` or `Functional.Try.Map_To_Result_With_Param`.

```
with Functional.Try.Map_To_Result_With_Param;

package body Infrastructure.Adapter.Console_Writer is
   -- Inner action may raise exceptions
   function Write_Action (Message : String) return Unit_Result.Result is
   begin
      Ada.Text_IO.Put_Line (Message);
      return Unit_Result.Ok (Unit_Value);
   end Write_Action;

   -- Error factory converts exception mapping into domain error
   function Make_Error (Kind : Error_Kind; Message : String)
     return Unit_Result.Result is
   begin
      return Unit_Result.Error (Kind, Message);
   end Make_Error;

   package Try_Write is new Functional.Try.Map_To_Result_With_Param (...);
   Mappings : constant Try_Write.Mapping_Array := [...];

   function Write (Message : String) return Unit_Result.Result is
```

```
   begin
      return Try_Write.Run (Message, Mappings);
   end Write;
end Infrastructure.Adapter.Console_Writer;
```

7.3.2. Prohibited Pattern: Manual Exception Handlers

Manual `exception` handlers in infrastructure adapters are forbidden because they create inconsistent error mapping, weaken auditability, and separate exception conversion from the shared policy.

7.3.3. Validation Enforcement

Release preparation and developer workflow validate these rules:

1. Infrastructure code shall use `Functional.Try` where boundary I/O exists.
2. Infrastructure code shall not contain manual `exception` handlers.
3. Domain/Application/API code shall not use the `exception` keyword.
4. Test code is exempt.

8. Build Configuration

8.1. GPR Projects

Project	Purpose
`hybrid_lib_ada.gpr`	Public library with `Library_Interface` restrictions.
`hybrid_lib_ada_internal.gpr`	Internal unrestricted project used for tests and examples.

8.2. Build Profiles

Profile	Target	Features
baremetal	Minimal runtime	Zero-footprint style profile.
embedded	Ravenscar embedded	Tasking-safe embedded profile.
standard	Desktop/server	Full feature set.

8.3. Library_Standalone Design Decision

hybrid_lib_ada uses:

```
Library_Standalone => "standard";
```

Rationale:

- hybrid_lib_ada is intended for composition within Ada toolchain ecosystems where the consumer supplies the Ada runtime at link time.
- Using `"encapsulated"` would bundle the full Ada runtime (RTS) into `libhybrid_lib_ada.a`. When two encapsulated stand-alone libraries (ESALs) are linked into the same executable, the linker emits thousands of `multiple definition` errors for the duplicated runtime symbols. This is a fundamental composition limitation of `"encapsulated"`, not a fixable defect.
- Public-API enforcement is achieved via `Library_Interface`, not via `Library_Standalone`. `Library_Interface` lists the packages clients may `with`; GPRbuild rejects any `with` clause naming an unlisted package. `Library_Standalone` is purely a packaging directive.

Therefore:

- `"standard"` is required for composability with other Ada libraries that may themselves be stand-alone.

- `"encapsulated"` is explicitly prohibited for this project.
- The architecture rule is enforced by `scripts/python/shared/arch_guard/adapters/ada.py` (`ROOT_GPR_ENCAPSULATED_ERROR`) on every `make check-arch`.

Note: `"encapsulated"` may be appropriate for standalone distribution to non-Ada environments (e.g., a C-callable library where the consumer has no Ada runtime), but this is not a requirement for hybrid_lib_ada.

9. Design Decisions

9.1. API.Operations as Child vs Sibling

Decision: use `API.Operations` as a child package instead of a sibling package.

Rationale: the child-package hierarchy is more idiomatic Ada, preserves a clean public API structure, and supports the design without losing SPARK value.

9.2. No Heap Allocation

Decision: use bounded strings and stack-oriented allocation strategies.

Rationale: this supports embedded compatibility, deterministic behavior, and SPARK-friendly design constraints.

9.3. Static vs Dynamic Polymorphism

Decision: use static polymorphism via generics.

Rationale: this supports SPARK compatibility, zero runtime overhead, and compile-time type safety.

10. Appendices

10.1. Package Dependency Notes

The package dependency structure reflects the same hybrid core as the application profile, with the API pattern providing the library-specific outer boundary. Supporting UML diagrams are maintained in `docs/diagrams/`.

10.2. Change History

Version	Date	Author	Changes
2.0.2	2026-04-24	Michael Gardner	Patch — added Library_Standalone Design Decision under Build Configuration. Documents the rationale for Library_Standalone = "standard" (composability across Ada toolchain ecosystems, avoidance of duplicate-runtime link failures from "encapsulated" stand-alone libraries) and credits Library_Interface as the public-API enforcement mechanism. Cites the arch_guard enforcement boundary (ROOT_GPR_ENCAPSULATED_ERROR).
2.0.1	2025-12-11	Michael Gardner	Added Section 6.3 Exception Boundary Specification with architecture diagram, layer rules, and required/prohibited patterns.
2.0.0	2025-12-09	Michael Gardner	Complete regeneration for v2.0.0; corrected Error_Kind (5 values); updated package structure.
1.0.0	2025-11-29	Michael Gardner	Initial release.